

Mini Project 2: Financial Ratios & DuPont Analysis

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1 Data Preparation and Winsorization

Following the assignment requirements, firm-level observations were extracted from the Compustat Fundamental Annual (*funda*) database for the five-year period from 2018 to 2022. The sample was restricted to standard industrial, domestic, and consolidated firm-year observations. In addition to the variables needed for the three-factor DuPont decomposition, I also kept two-digit historical SIC codes, the market-to-book ratio, and other firm characteristics such as liquidity, solvency, and firm size.

After applying the screening conditions, the final sample contains 31,360 firm-year observations and 8,106 distinct firms. Since raw accounting data often contain extreme values caused by very small denominators, unusual reporting outcomes, or large temporary losses, winsorization was necessary before conducting the main analysis. Table 1 compares selected summary statistics before and after applying 1% and 99% winsorization in SAS.

As shown in Panel A, the raw data include several extreme and economically unrealistic values. For example, the maximum ROE reaches 1000.00, Profit Margin reaches 2033.00, Total Asset Turnover reaches 514000.00, and the Equity Multiplier reaches 29586.74. After winsorization in Panel B, these extreme values were capped at the upper and lower tails, while the medians remained unchanged. This suggests that the procedure worked as intended: it reduced the influence of outliers without removing observations or materially changing the center of the distributions.

Table 1: Selected Summary Statistics Before and After Winsorization

Variable	N	Mean	Median	Min	Max
<i>Panel A: BEFORE Winsorization</i>					
ROE	31,360	-1.4605	0.0570	-1801.05	1000.00
PM	31,360	-14.0708	0.0416	-84221.67	2033.00
TAT	31,360	16.9993	0.4143	0.0000	514000.00
EM	31,360	6.5368	2.3831	1.0000	29586.74
<i>Panel B: AFTER 1% and 99% Winsorization</i>					
ROE	31,360	-0.1966	0.0570	-8.0468	1.3839
PM	31,360	-2.6733	0.0416	-60.8315	1.4461
TAT	31,360	0.5884	0.4143	0.0023	3.0786
EM	31,360	4.2243	2.3831	1.0335	33.9270

Table 2 reports the mean and median values of ROE and the three DuPont components after winsorization. The mean values of ROE and Profit Margin are much lower than their medians,

indicating that the sample is still left-skewed even after controlling for extreme outliers. By contrast, Total Asset Turnover remains positive and relatively stable, while the Equity Multiplier stays well above 1 on average, showing that leverage remains an important part of the ROE structure.

Table 2: Mean and Median of ROE and DuPont Components After Winsorization

Variable	Mean	Median
ROE	-0.1966	0.0570
PM	-2.6733	0.0416
TAT	0.5884	0.4143
EM	4.2243	2.3831

2 Trend of DuPont 3 Factors (2018–2022)

The DuPont framework decomposes ROE into three drivers: Profit Margin, Total Asset Turnover, and the Equity Multiplier. Figures 1 to 3 present the five-year trends of the three DuPont components, while Figure 4 shows the corresponding movement in ROE. Table 3 reports the yearly mean and median values.

Table 3: Yearly Mean and Median of ROE and DuPont Components

Year	ROE Mean	PM Mean	TAT Mean	EM Mean	ROE Median	PM Median	TAT M
2018	-0.1577	-2.1751	0.6443	4.0295	0.0656	0.0475	
2019	-0.2144	-2.3449	0.6005	4.2432	0.0551	0.0394	
2020	-0.2404	-2.8067	0.5357	4.3218	0.0289	0.0243	
2021	-0.1342	-2.9767	0.5498	4.0792	0.0718	0.0536	
2022	-0.2380	-3.0398	0.6130	4.4483	0.0618	0.0425	

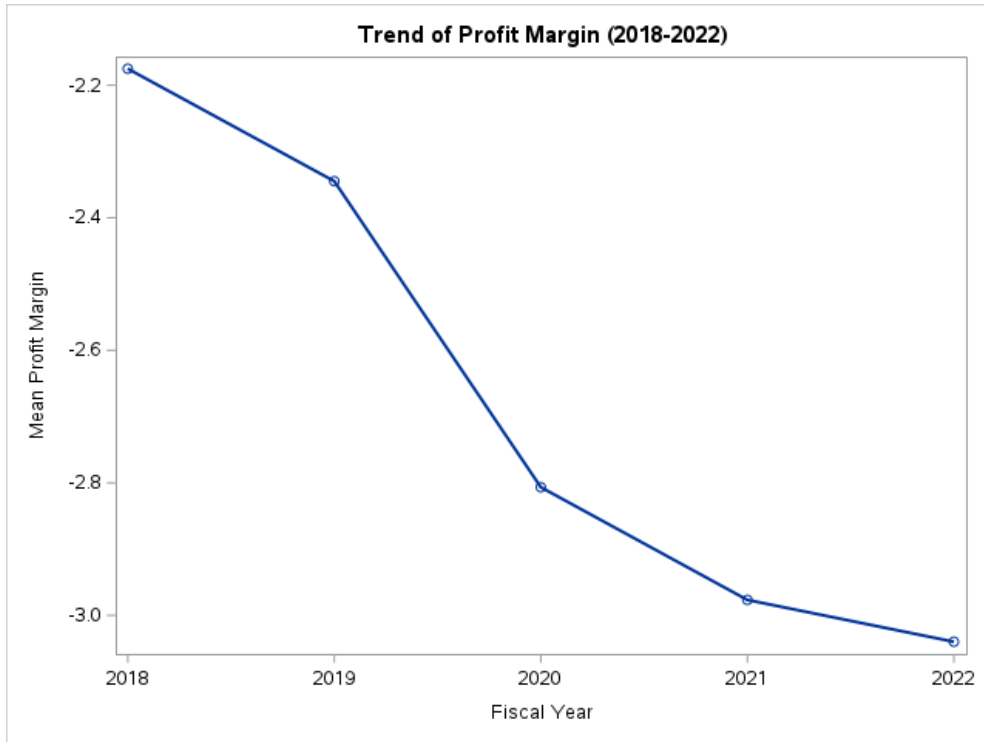


Figure 1: Trend of Profit Margin (2018–2022)

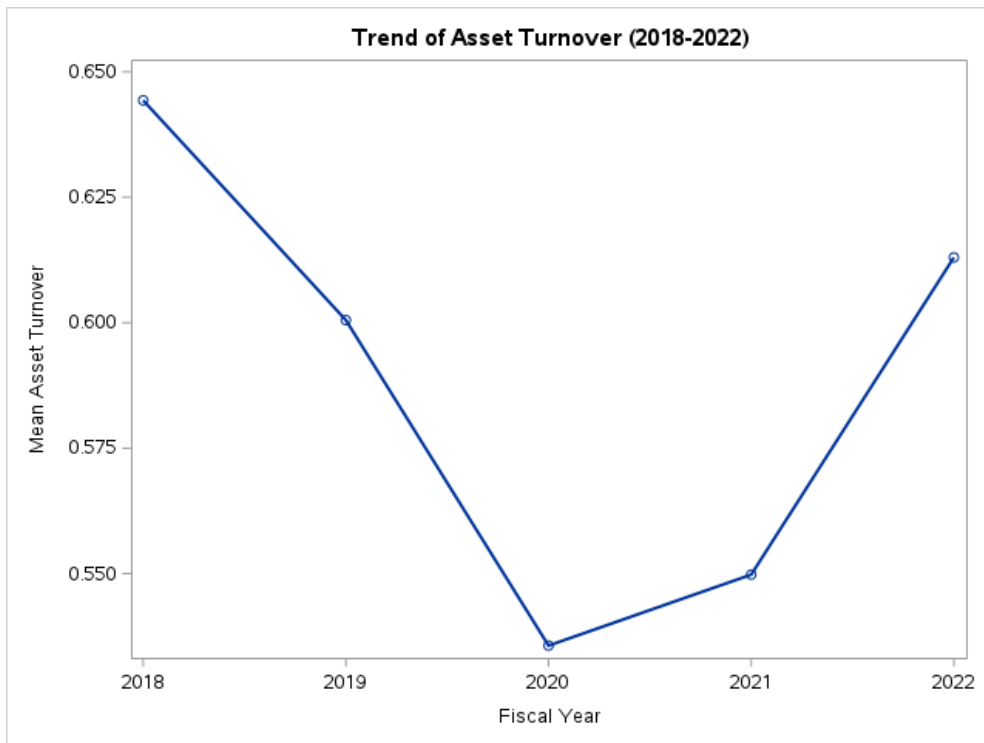


Figure 2: Trend of Asset Turnover (2018–2022)

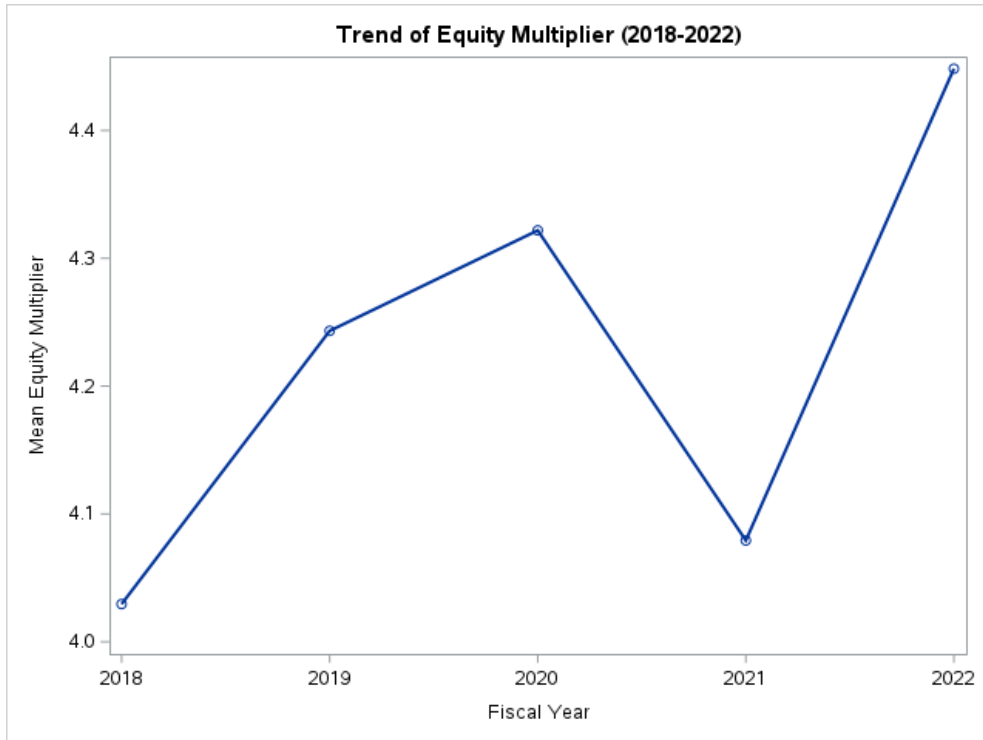


Figure 3: Trend of Equity Multiplier (2018–2022)

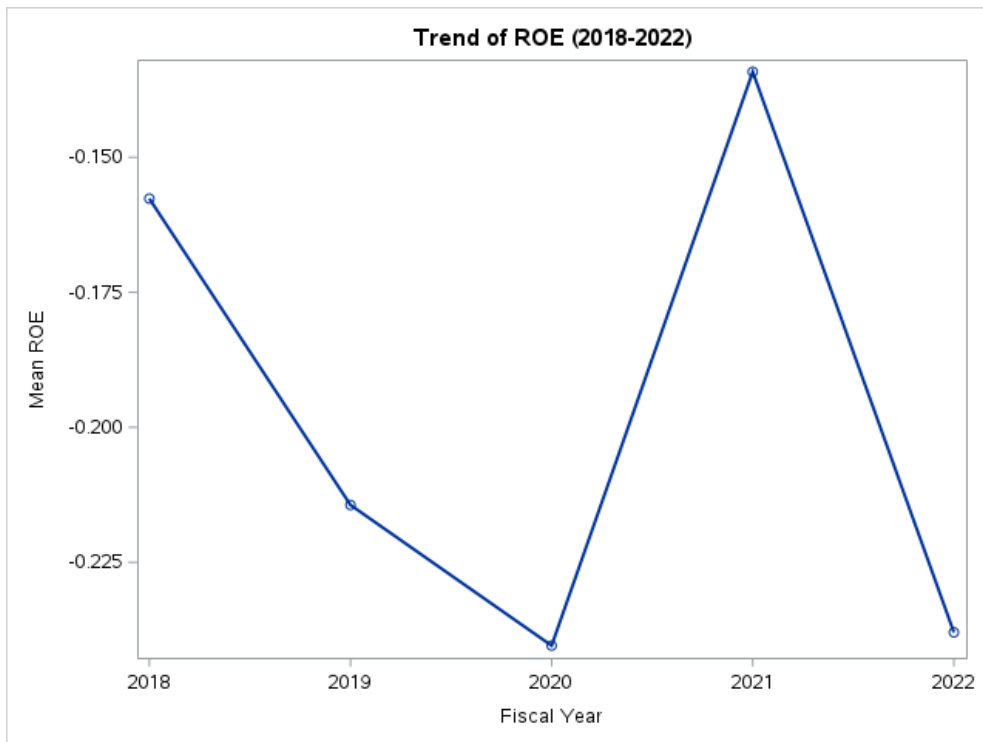


Figure 4: Trend of ROE (2018–2022)

The trend suggests a fairly intuitive story. Profit Margin becomes more negative over time and remains especially weak after 2020, indicating that profitability is the main factor dragging down average ROE. Total Asset Turnover drops noticeably in 2020, which is consistent with weaker asset-use efficiency during that year, and then partially recovers in 2021 and 2022. At

the same time, the Equity Multiplier stays relatively high throughout the sample and rises further in 2022, suggesting that firms relied more on leverage while profitability remained under pressure.

The ROE series broadly reflects the same pattern. ROE worsens from 2018 to 2020, improves in 2021, and then declines again in 2022. This fluctuation is consistent with the idea that the modest recovery in turnover after 2020 was not enough to fully offset the continuing weakness in profit margins. In other words, the overall weakness in ROE during this period appears to be driven more by poor profitability than by a permanent collapse in operating efficiency. The increase in the Equity Multiplier also implies that some firms were using more leverage to support returns.

3 Determinants of ROE: Regression Analysis

To provide some additional evidence, I also estimated an OLS regression of ROE on the three DuPont components and several firm characteristics, including the Current Ratio, Debt-to-Equity ratio, Market-to-Book ratio, and firm size. Although regression analysis was not explicitly required in the assignment, it helps show whether these relationships remain statistically meaningful after controlling for other firm-level differences.

As shown in Table 4, the model is highly significant overall, with an F -statistic of 1607.00 and an R^2 of 0.3352. Profit Margin, Total Asset Turnover, and the Equity Multiplier all have positive and statistically significant coefficients, which is broadly consistent with the DuPont framework. The Current Ratio is also positively related to ROE, suggesting that firms with stronger short-term liquidity tend to perform better in this sample. In contrast, the Debt-to-Equity ratio is significantly negative, indicating that a heavier debt burden is associated with weaker shareholder returns once the other variables are held constant. Firm size is positively associated with ROE, while the coefficient on MTB is negative in this specification.

Table 4: OLS Regression Results for ROE

Variable	Estimate	Standard Error	t Value	Pr > t
Intercept	-1.5003	0.0276	-54.34	< .0001
Profit Margin (PM)	0.0122	0.0004	30.71	< .0001
Asset Turnover (TAT)	0.2819	0.0105	26.84	< .0001
Equity Multiplier (EM)	0.1192	0.0151	7.90	< .0001
Current Ratio (CR)	0.0337	0.0016	21.36	< .0001
Debt to Equity (DE)	-0.2140	0.0154	-13.85	< .0001
Market to Book (MTB)	-0.0073	0.0008	-9.45	< .0001
Size	0.1659	0.0027	62.35	< .0001
<i>Model Fit: F-Value = 1607.00, R-Square = 0.3352, p-value < 0.0001</i>				

4 Conclusion

This mini-project uses SAS to clean, winsorize, and analyze a large Compustat firm-year sample through the DuPont framework. The main finding is that the weakness in average ROE over 2018–2022 is primarily related to poor profitability. Asset turnover declines in 2020 but later

shows some recovery, while leverage remains high and rises again by 2022. The regression results support the same overall interpretation by showing that profitability, turnover, and leverage all remain important determinants of ROE after controlling for liquidity, solvency, market valuation, and size.

In short, the decline in shareholder returns during this period seems to be more of a profitability problem than a pure efficiency problem. Higher leverage may have helped cushion the pressure for some firms, but it was not enough to fully restore average ROE.